6.2.7 The DNA polymerase

The DNA polymerase or the DNA dependent DNA synthetase is a large multi subunit enzyme which adds deoxyribonucleotides to the 3' end of the growing strand one by one. Hence it could be thought of an enzyme that elongates the existing nucleic acid (DNA or RNA primer) in 5'->3' direction. The DNA polymerase enzyme can perform two other activities which are the 3'→5' exonuclease activity and 5'→3' exonuclease activity. Exonucleases are enzymes which can remove bases one by one from the ends of a DNA molecule. The 3'→5' exonuclease activity of DNA polymerases provides as an error correction or proof reading action by removing incorrectly incorporated bases during replication. There are different types of DNA polymerases adapted to different types of cells/organisms and the features described above are of the most predominate enzyme called DNA polymerase I. There are other DNA polymerases like DNA polymerase II, III, alpha, beta, delta etc.,

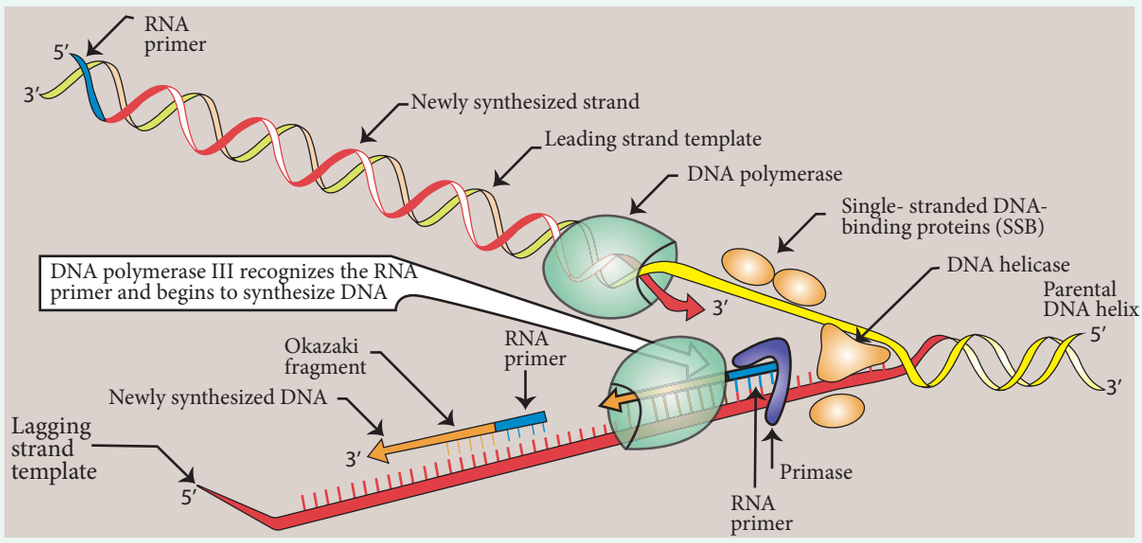


Figure 4 Overview of the DNA replication process

6.2.8 Difference between prokaryotes and eukaryotes in DNA replication

S. No.	Prokaryotes	Eukaryotes
1	It occurs inside the cytoplasm	It occurs inside the nucleus
2	There is a single origin of replication	Origin of replication are numerous
3	DNA polymerase III carries out both initiation and elongation	Initiation is carried out by DNA polymerase alpha while elongation by DNA polymerase by delta and epsilon
4	DNA repair and gap filling are done by polymerase I	The same are performed by DNA polymerase beta

XII U6 Mol Bio.indd 144 4/8/2020 3:16:24 PM